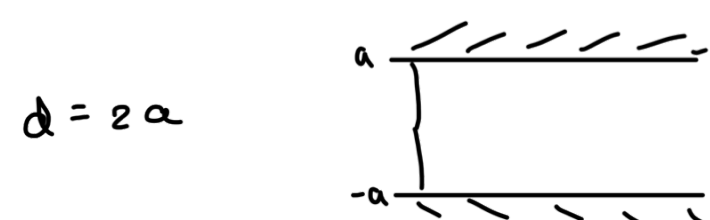


1.

$$E_x(y, z) = \sum_m a_m U_m(y) e^{-i\beta_m z} \quad U_m(y) = \begin{cases} \sqrt{\frac{2}{d}} \cos \frac{m\pi y}{d} & , m = 1, 3, 5, \dots \\ \sqrt{\frac{2}{d}} \sin \frac{m\pi y}{d} & , m = 2, 4, 6, \dots \end{cases}$$



$$E_0 = E_x(0, 0) \Rightarrow E_0 = a_1 U_1(y) e^{-i\beta_1 z} \Rightarrow E_0 = a_1 \sqrt{\frac{2}{2a}} \cos\left(\frac{\pi y}{2a}\right) e^{-i\beta_1 z} \Rightarrow E_0 = a_1 \sqrt{\frac{2}{a}} \Rightarrow a_1 = E_0 \sqrt{a}$$

$$P_z = \frac{1}{2\eta} |E_0 \sqrt{a}|^2 \cos \theta_1 = \frac{1}{2\eta} |E_0|^2 a \cos \theta_1$$

$$\beta_1 = k \cos \theta_1 \Rightarrow \cos \theta_1 = \frac{\beta_1}{k} \Rightarrow \cos^2 \theta_1 = \frac{\beta_1^2}{k^2} \Rightarrow \cos^2 \theta_1 = \frac{k^2}{k^2} - m^2 \frac{\pi^2}{d^2} \frac{1}{k^2} \Rightarrow \cos^2 \theta_1 = 1 - m^2 \frac{\pi^2}{d^2} \frac{\lambda^2}{4\pi^2} \Rightarrow \cos \theta_1 = \sqrt{1 - \frac{m^2 \lambda^2}{4a^2 \cdot 4}} = \sqrt{1 - \frac{\lambda^2}{16a^2}}$$

$$\beta_m^2 = k^2 \cos^2 \theta_m \Rightarrow \beta_m^2 = k^2 (1 - \sin^2 \theta_m) \Rightarrow \beta_m^2 = k^2 - \frac{4\pi^2}{\lambda^2} \cdot \frac{m^2 \lambda^2}{4d^2} \Rightarrow \beta_m^2 = k^2 - m^2 \frac{\pi^2}{d^2}$$

$$2a \leq \lambda < 4a$$

$$\cos \theta_1 = \sqrt{1 - \frac{\lambda^2}{16a^2}} = \begin{cases} \sqrt{1 - \frac{4a^2}{16a^2}} & \Rightarrow \sqrt{1 - \frac{1}{4}} \\ \sqrt{1 - \frac{16a^2}{16a^2}} & \Rightarrow \sqrt{1 - 1} \end{cases} \Rightarrow \begin{cases} \sqrt{\frac{3}{4}} & \text{para } \lambda = 2a \\ 0 & \text{para } \lambda = 4a \end{cases}$$

Para $\lambda = 2a$:

$$P_z = \frac{1}{2\eta} |E_0|^2 a \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2.2\eta} |E_0|^2 a \quad 0 < P_z \leq \frac{\sqrt{3}}{4\eta} |E_0|^2 a$$

Para $\lambda = 4a$:

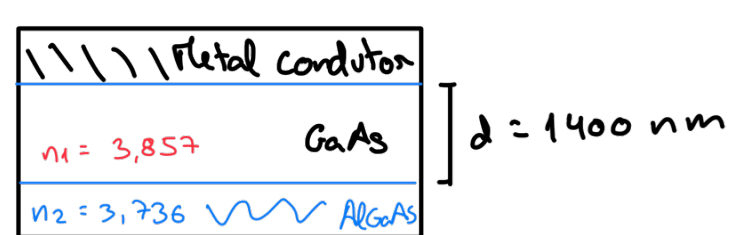
$$P_z = \frac{1}{2\eta} |E_0|^2 a \times 0 = 0$$

Para $\lambda = 4a$:

$$\Gamma = \frac{2d}{\lambda} = \frac{2 \cdot 2a}{4a} = 1 \neq 0, \text{ logo para } \lambda = 4a \text{ não há modos guiados, já que } \Gamma = 0$$

Na banda sem modos guiados não há potência porque não há transporte de energia, logo para $\lambda = 4a$ como não tem modos guiados não há potência

2. a) $\lambda_0 = 632,8 \text{ nm}$ $d = 1400 \text{ nm}$ $n = 3,857$



$$\sin \theta_a = n_1 \sin \bar{\theta}_c \Rightarrow \sin 74^\circ = 3,857 \sin \bar{\theta}_c \Rightarrow \sin \bar{\theta}_c = 0,25 \Rightarrow \bar{\theta}_c = 14,45^\circ$$

$$\cos \bar{\theta}_c = \frac{n_2}{n_1} \Rightarrow \bar{\theta}_c = \arccos\left(\frac{n_2}{n_1}\right) \Rightarrow \bar{\theta}_c = 14,39^\circ \approx 14,45^\circ$$

b) $NA = \sqrt{n_1^2 - n_2^2} = \sqrt{3,857^2 - 3,736^2} = 1,17$

$$NA_{10,8\%} = \sqrt{3,857^2 - 3,736^2} = 0,95$$

$$\sin \theta_a = NA$$

$$-1 \leq \sin \theta \leq 1$$

$NA_{10,8\%} < 1$, θ_a é real e a guia pode capturar e conduzir luz

$NA_{31,5\%} > 1$, θ_a é imaginário, logo não captura luz

c) i) GaAs $\rightarrow$ AlGaAs: $\tan\left(\frac{\varphi_2}{2}\right) = \frac{\sin^2 \bar{\theta}_c \cdot 1}{\sin^2 \theta}$

$$\frac{2\pi}{\lambda} (2d \sin \theta) - \varphi_1 - \varphi_2 = 2m\pi$$

$$\Rightarrow \frac{2\pi}{\lambda} 2d \sin \theta - \pi - \varphi_2 = 2m\pi$$

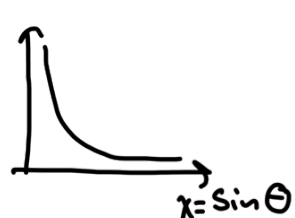
$$\Rightarrow \frac{2\pi d}{\lambda} \sin \theta - \frac{\pi}{2} - \frac{\varphi_2}{2} = m\pi$$

$$\Rightarrow \frac{2\pi d}{\lambda} \sin \theta - m\pi - \frac{\pi}{2} = \frac{\varphi_2}{2}$$

$$\Rightarrow \tan\left(\frac{2\pi d}{\lambda} \sin \theta - m\pi - \frac{\pi}{2}\right) = \tan\left(\frac{\varphi_2}{2}\right)$$

$$\Rightarrow -\cot\left(\frac{2\pi d}{\lambda} \sin \theta - m\pi\right) = \frac{\sin^2 \bar{\theta}_c}{\sin^2 \theta} \cdot 1$$

ii) $f(x) = \frac{\sin^2 \bar{\theta}_c}{\sin^2 \theta} \cdot 1$: $x \rightarrow 0 \Rightarrow f(x) \rightarrow +\infty$
 $x \rightarrow \sin \bar{\theta}_c \Rightarrow f(x) \rightarrow 0$

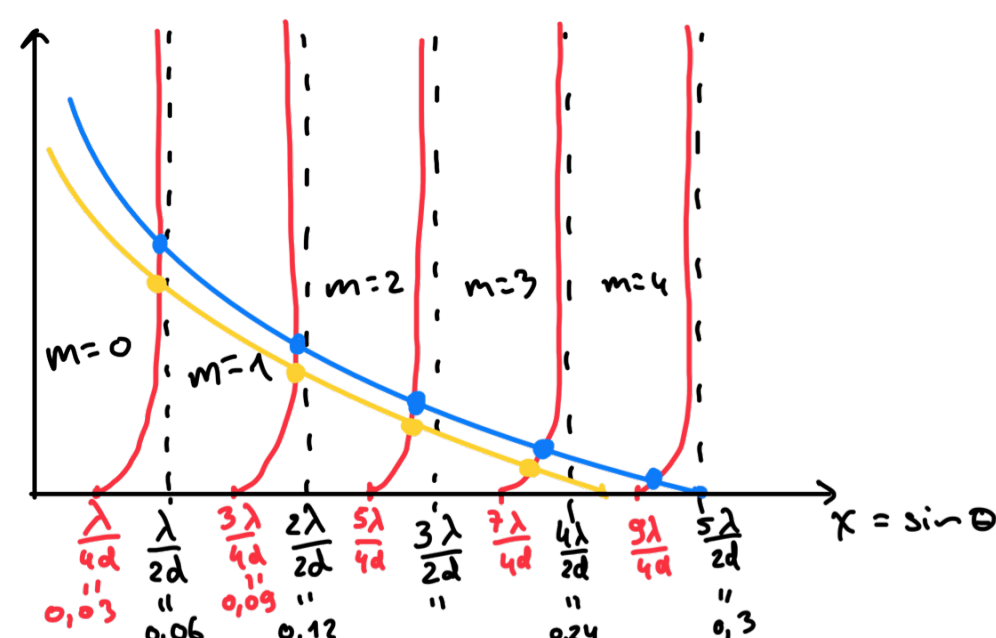


Para m par e ímpar:

$$f(x) = -\cot\left(\frac{2\pi d}{\lambda} \sin \theta - m\pi\right) = -\cot\left(\frac{2\pi d}{\lambda} \sin \theta\right) : \quad m = 2n+1$$

$$\text{zeros: } -\cot\left(\frac{2\pi d}{\lambda} x\right) = 0 \Rightarrow \frac{2\pi d}{\lambda} x = \frac{\pi}{2} + n\pi \Rightarrow x = \left(\frac{1}{2} + n\right) \frac{\lambda}{2d} \Rightarrow x = (2n+1) \frac{\lambda}{4d} = m \frac{\lambda}{4d}$$

$$\text{extremos: } -\cot\left(\frac{2\pi d}{\lambda} x\right) = \pm \infty \Rightarrow \frac{2\pi d}{\lambda} x = n\pi \Rightarrow x = n \frac{\lambda}{2d} \Rightarrow x = (2n+2) \frac{\lambda}{2d} = (n+1) \frac{\lambda}{d}$$



$$\text{Al-31,5\%: } f(x) = \sqrt{\frac{0,09}{x^2} - 1}$$

$$\sin \theta_a = n_1 \sin \bar{\theta}_c \Rightarrow \sin \bar{\theta}_c = \frac{NA}{n_1} = 0,30$$

$$\text{Al-19,8\%: } f(x) = \sqrt{\frac{0,0625}{x^2} - 1}$$

$$\frac{\lambda}{2d} = \frac{\lambda_0}{2n_1 d} = \frac{632,8 \text{ nm}}{2 \times 3,857 \times 1400 \text{ nm}} = 0,06$$

$$\frac{2\lambda}{2d} = \frac{2\lambda_0}{2n_1 d} = 0,117$$

$$\frac{\lambda}{4d} = \frac{\lambda_0}{4n_1 d} = 0,029$$

$$\frac{2\lambda}{4d} = \frac{2\lambda_0}{4n_1 d} = 0,06$$

iii)

iv) Para $x < \frac{\lambda}{4d}$ não há modos:

$$\sin \bar{\theta}_c = \frac{\lambda_c}{4d} \Rightarrow \lambda_c = 4d \sin \bar{\theta}_c \Rightarrow \frac{2\pi \omega_c}{\omega_c} = 4d \sin \bar{\theta}_c \Rightarrow \omega_c = \frac{\pi c}{2d \sin \bar{\theta}_c} \xrightarrow{\text{para } \sin \bar{\theta}_c = 1} \omega_c = \frac{\pi c}{2d}$$

$$\text{Ressonâncias: } \omega_c = \frac{\pi c}{d} \quad \text{pk } \sin \bar{\theta}_c = 1 \text{ e surge no caso de 2 espelhos}$$

como neste caso só temos 1 espelho, então $\omega_c = \frac{\pi c}{2d}$

v) $\frac{\lambda}{4d} < \sin \bar{\theta}_c < \frac{\lambda}{2d}$

$$\sin \theta_a = n_1 \sin \bar{\theta}_c \Rightarrow \sin \bar{\theta}_c = \frac{NA}{n_1}$$

$$\Rightarrow \frac{\lambda}{4d} < \frac{NA}{n_1} < \frac{\lambda}{2d}$$

$$\Rightarrow \frac{\lambda_0}{4n_1 d} < \frac{NA}{n_1} < \frac{\lambda_0}{2n_1 d}$$

$$\Rightarrow \frac{\lambda_0}{4d} < NA < \frac{\lambda_0}{2d} \Rightarrow \frac{1}{2} < \frac{2d}{\lambda_0} NA < 1$$